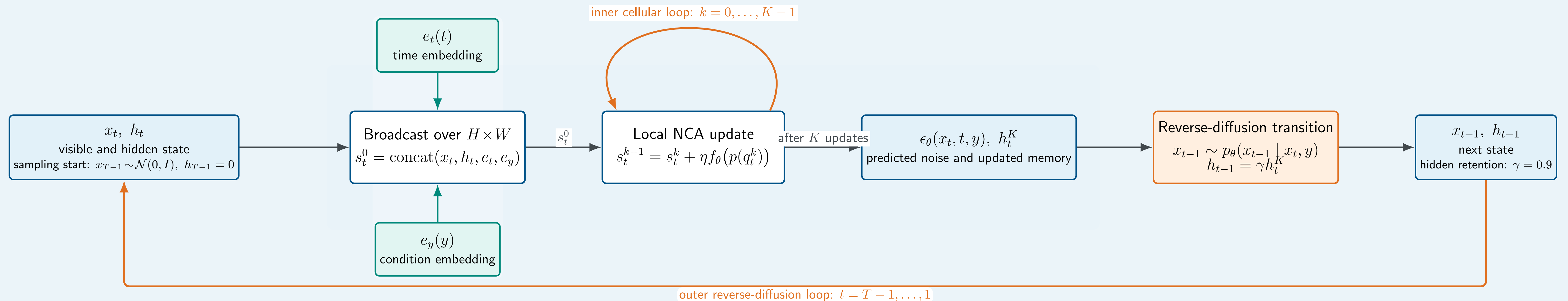


Pixel-Space Diffusion with Neural Cellular Automata

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Diffusion-NCA Sampling Pipeline



Abstract

We use a neural cellular automaton (NCA) as the noise predictor of a denoising diffusion model. At each diffusion level, the NCA performs a short sequence of updates, avoiding backpropagation through the complete generation trajectory. This separates diffusion time, which determines corruption, from cellular time, which determines local computation. A class-conditioned model generates recognizable and diverse MNIST digits, while exploratory sweeps reveal a trade-off between diffusion depth, cellular computation, and sample quality.

Introduction

Neural Cellular Automata (NCA) learn spatially shared local rules acting on visible and hidden cell states. One major, long-standing issue with NCA are scaling them to run over longer time horizons due to an explosion of VRAM requirements as well as vanishing/exploding gradients. We propose Diffusion neural cellular automata. Diffusion models learn to reverse a fixed Gaussian corruption process (Sohl-Dickstein et al., 2015; Ho et al., 2020). Their denoiser is normally a U-Net or another multiscale network (Ronneberger et al., 2015), but we replace this denoiser with an NCA that performs K updates at each of T diffusion levels. Training backpropagates through the K cellular updates, but not through the full reverse trajectory.

Diffusion NCA

Let $x_0 \in [-1, 1]^{C \times H \times W}$ be a clean image. We use zero-based indices $t \in \{0, \dots, T-1\}$ and define $\bar{\alpha}_{-1} = 1$. With $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{j=0}^t \alpha_j$, the forward process is

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (1)$$

We use a cosine schedule (Nichol and Dhariwal, 2021) with offset $s = 0.008$ and clip β_t to $[10^{-5}, 0.02]$.

The cellular state is $s = [x, h]$, where x is the visible channel and $h \in \mathbb{R}^{C_h \times H \times W}$ contains hidden channels. Every cell receives a learned diffusion-time embedding and an eight-channel class embedding.

$$s_t^{k+1} = s_t^k + \eta f_\theta(p(q_t^k)), \quad k = 0, \dots, K-1, \quad (2)$$

where $\eta = 0.15$. Individual updates are local. We make sure to not include any positional encoding, spatial statistics like Groupnorm and use reflective padding.

The visible state is initialized with x_t and interpreted after K updates as the predicted noise $\epsilon_\theta(x_t, t, y)$. Training minimizes

$$\mathcal{L}_\epsilon = \mathbb{E}_{x_0, t, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t, y)\|_2^2 \right]. \quad (3)$$

Only ordinary class-conditioned generation is evaluated here. Hidden activations receive a small overflow penalty, and all reported samples use exponential-moving-average parameters with decay 0.999.

Hidden-State Representations during inference

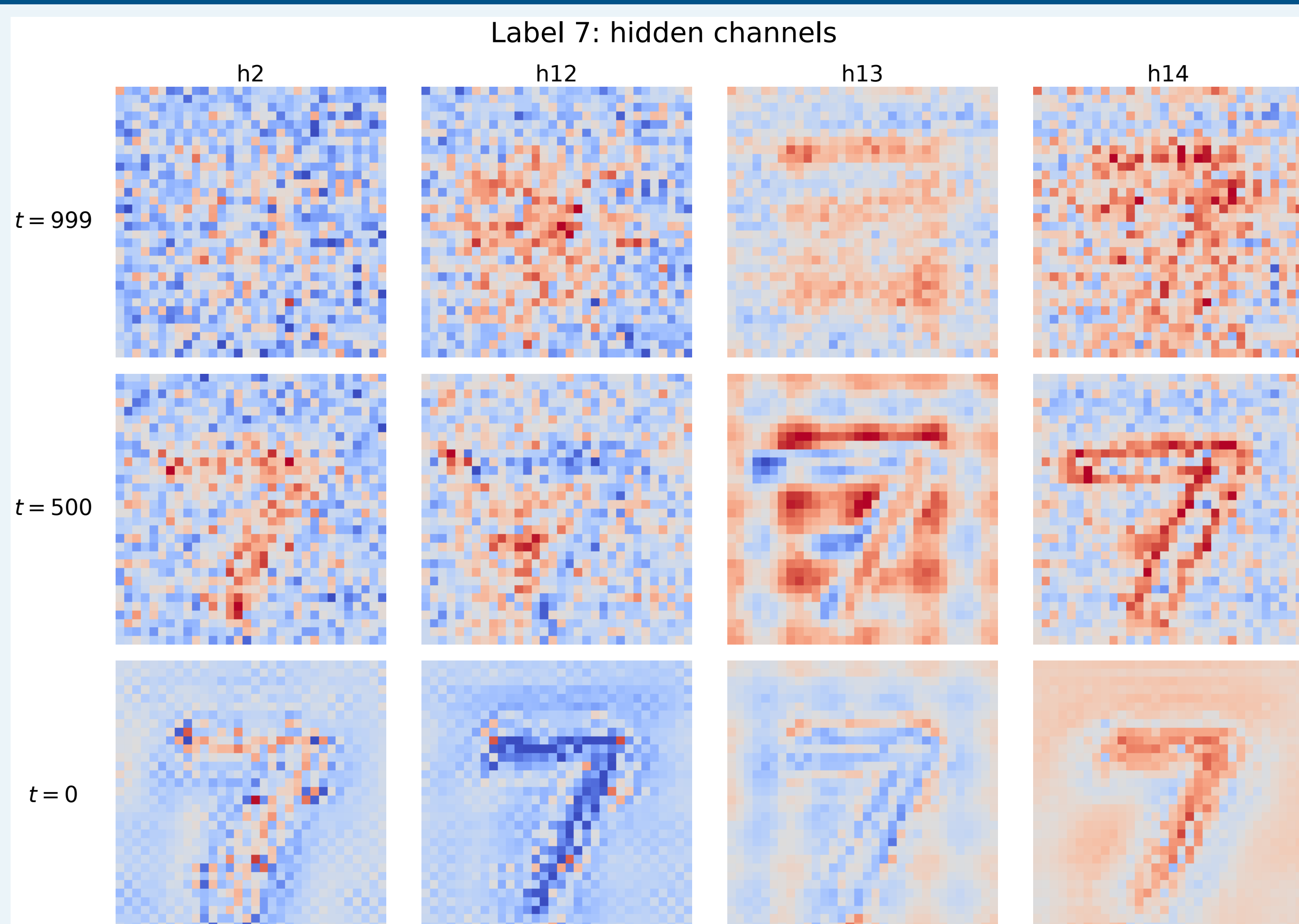


Figure 1: Four hidden channels for a class-7 sample at early, middle, and final reverse-diffusion levels.

Experiments

We train on MNIST (LeCun et al., 1998), resized and padded to 32×32 , using AdamW (Loshchilov and Hutter, 2019). The strongest model has 32 hidden channels, reaction width 128, $T = 1000$, $K = 8$, batch size 256, learning rate 2×10^{-4} , weight decay 10^{-6} , and 276,553 trainable parameters.

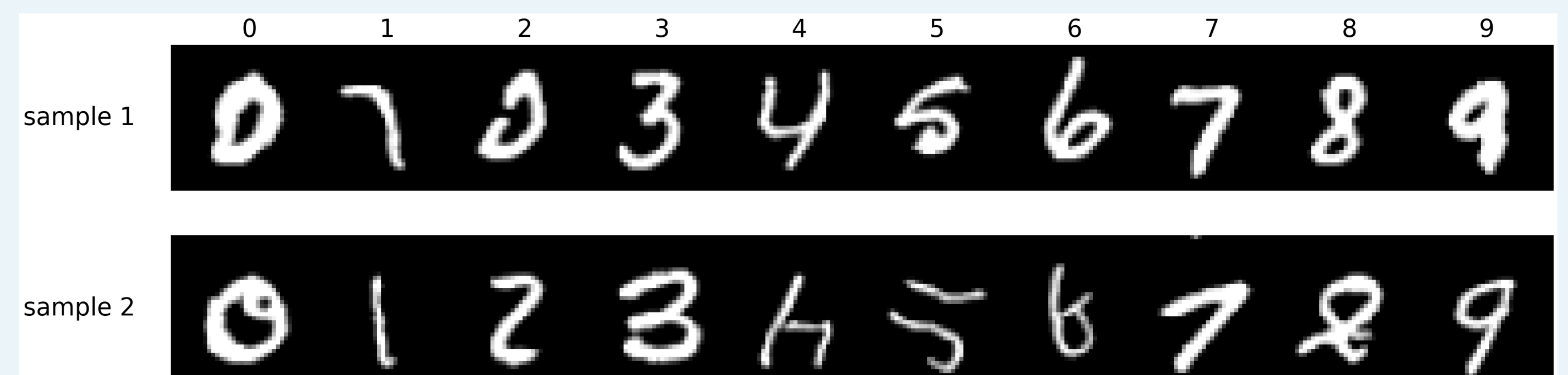


Figure 2: Unfiltered class-conditioned samples. The requested class is shown above each column.

Figure 2 shows recognizable examples from all classes, with noticeable diversity, although there are some expectations.

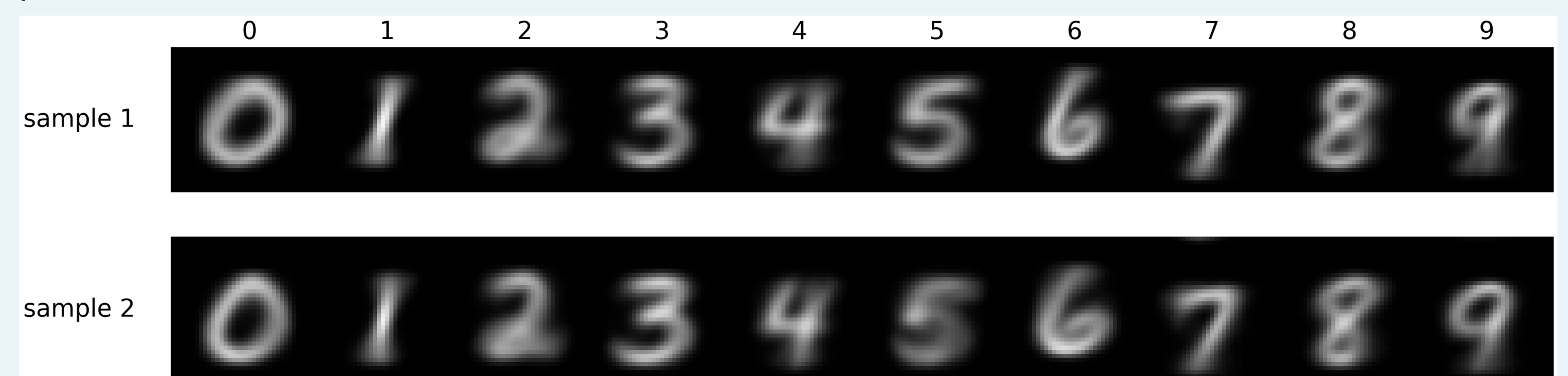


Figure 3: Conditional samples from the NCA baseline (no reverse diffusion). The requested class is shown above each column.

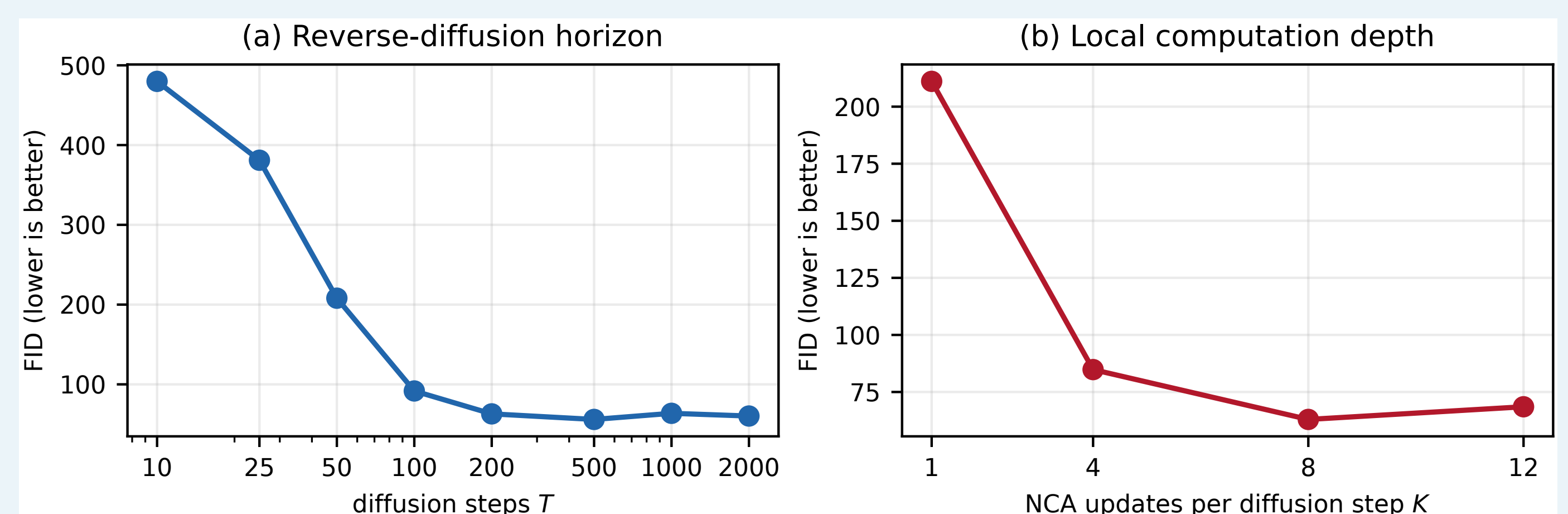


Figure 4: Exploratory FID sweeps over diffusion steps T and cellular updates K . Each point uses 128 real and generated images.

Figure Figure 4 shows that the FID improves with more diffusion and with more cellular steps, however we also see diminishing returns.

Discussion and Conclusion

The experiments establish that an NCA can act as a pixel-space diffusion denoiser. Standard NCA training propagates a loss through a long developmental trajectory, which is infeasible for long trajectories, where as our diffusion training can split this up by sampling t uniformly during training and during inference sample through the full trajectory.

In future work we plan to incorporate techniques such such as DDIM (Song et al., 2021), test on larger, more complicated images and perform interventions on the hidden states. Overall, the results provide a route toward generative cellular systems that learn from corrupted states, support class-conditioned diversity, and retain local spatial interactions.

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